Q1 - 24 June - Shift 1

A metre scale is balanced on a knife edge at its centre. When two coins, each of mass 10 g are put one on the top of the other at the 10.0 cm mark the scale is found to be balanced at 40.0 cm mark. The mass of the metre scale is found to be $x \times 10^{-2}$ kg. The value of x is

Space for your notes:

Q2 - 24 June - Shift 2

A fly wheel is accelerated uniformly from rest and rotates through 5 rad in the first second. The angle rotated by the fly wheel in the next second, will be :

(A) 7.5 rad
(B) 15 rad
(C) 20 rad
(D) 30 rad

Space for your notes:

Q3 - 25 June - Shift 1

If force $\vec{F} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ acts on a particle having position vector $2\hat{i} + \hat{j} + 2\hat{k}$ then, the torque about the origin will be :-

(A) $3\hat{i} + 4\hat{j} - 2\hat{k}$

(B) $-10\hat{i} + 10\hat{j} + 5\hat{k}$

(C) $10\hat{i} + 5\hat{j} - 10\hat{k}$

(D) $10\hat{i} + \hat{j} - 5\hat{k}$

Space for your notes:

Q4 - 25 June - Shift 2

Moment of Inertia (M.I.) of four bodies having same mass 'M' and radius '2R' are as follows:

I_1 = M.I. of solid sphere about its diameter

I_2 = M.I. of solid cylinder about its axis

I_3 = M.I. of solid circular disc about its diameter

I_4 = M.I. of thin circular ring about its diameter

If $2(I_2 + I_3) + I_4 = x.\ I_1$ then the value of x will be _____

Space for your notes:

Q5 - 26 June - Shift 1

A thin circular ring of mass M and radius R is rotating with a constant angular velocity 2 rads^{-1} in a horizontal plane about an axis vertical to its plane and passing through the center of the ring. If two objects each of mass m be attached gently to the opposite ends of a diameter of ring, the ring will then rotate with an angular velocity (in rads^{-1}).

(A) $\frac{M}{(M+m)}$ (B) $\frac{(M+2m)}{2M}$

(C) $\frac{2M}{(M+2m)}$ (D) $\frac{2(M+2m)}{M}$

Space for your notes:

Q6 - 26 June - Shift 2

Solid spherical ball is rolling on a frictionless horizontal plane surface about its axis of symmetry. The ratio of rotational kinetic energy of the ball to its total kinetic energy is :-

(A) $\frac{2}{5}$ (B) $\frac{2}{7}$ (C) $\frac{1}{5}$ (D) $\frac{7}{10}$

Space for your notes:

Q7 - 28 June - Shift 1

Match List-I with List-II

	List-I		List-II
A	Moment of inertia of solid sphere of radius R about any tangent	I	$\frac{5}{3}MR^2$
B	Moment of inertia of hollow sphere of radius (R) about any tangent	II	$\frac{7}{5}MR^2$
C	Moment of inertia of circular ring of radius (R) about its diameter.	III	$\frac{1}{4}MR^2$
D	Moment of inertia of circular disc of radius (R) about any diameter.	IV	$\frac{1}{2}MR^2$

Question: Choose the correct answer from the options given below

(A) A-II, B-II, C-IV, D-III

(B) A-I, B-II, C-IV, D-III

(C) A-II, B-I, C-III, D-IV

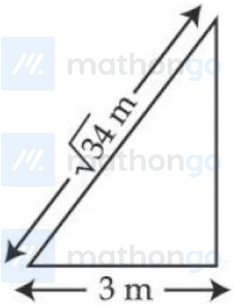
(D) A-I, B-II, C-III, D-IV

Space for your notes:

Q8 - 28 June - Shift 2

A $\sqrt{34}$ m long ladder weighing 10 kg leans on a frictionless wall. Its feet rest on the floor 3 m away from the wall as shown in the figure. If F_f and F_w are the reaction forces of the floor and the wall, then ratio of F_w/F_f will be:

(Use $g = 10\ m/s^2$)



(A) $\frac{6}{\sqrt{110}}$ (B) $\frac{3}{\sqrt{113}}$

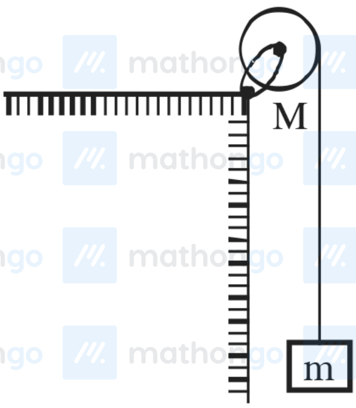
(C) $\frac{3}{\sqrt{109}}$ (D) $\frac{2}{\sqrt{109}}$

Space for your notes:

Q9 - 28 June - Shift 2

A uniform disc with mass M = 4 kg and radius R = 10 cm is mounted on a fixed horizontal axle as shown in figure. A block with mass m = 2 kg hangs from a massless cord that is wrapped around the rim of the disc. During the fall of the block, the cord does not slip and there is no friction at the axle. The tension in the cord is ________N.

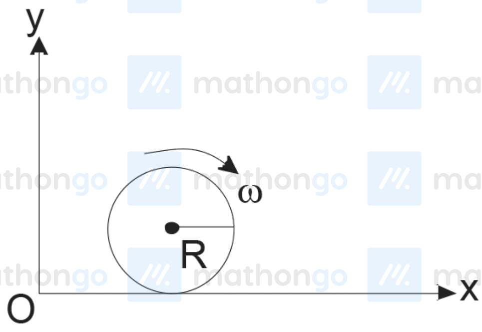
(Take $g = 10\ ms^{-2}$)



Space for your notes:

Q10 - 29 June - Shift 1

A spherical shell of 1 kg mass and radius R is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin O is $\frac{a}{3}R^2\omega$. The value of a will be :



(A) 2 (B) 3

(C) 5 (D) 4

Space for your notes:

Q11 - 29 June - Shift 2

The moment of inertia of a uniform thin rod about a perpendicular axis passing through one end is I_1. The same rod is bent into a ring and its moment of inertia about a diameter is I_2. If $\frac{I_1}{I_2}$ is $\frac{x\pi^2}{3}$, then the value of x will be________.

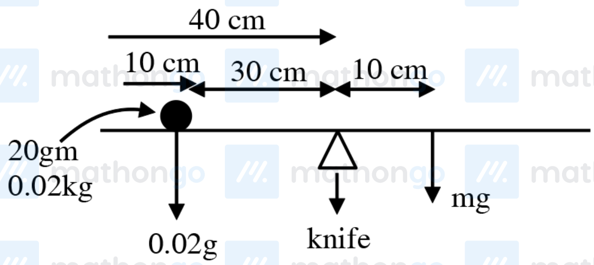
Space for your notes:

Answer Key

Q1 (6)	**Q2** (B)	**Q3** (B)	**Q4** (5)
Q5 (C)	**Q6** (B)	**Q7** (A)	**Q8** (C)
Q9 (10)	**Q10** (C)	**Q11** (8)	

Q1 (6)

Let mass of meter scale be m.



Balancing torque about knife edge

$(0.02g) \times (30 \times 10^{-2}) = mg \times (10 \times 10^{-2})$

$m = 0.06 \text{ kg} = 6 \times 10^{-2} \text{ kg}$

Q2 (B)

$$5 = \frac{1}{2}\alpha(1)^2$$

$$\theta = \frac{1}{2}\alpha(2)^2$$

$$\theta - 5 = 15$$

Q3 (B)

$\vec{\tau} = \vec{r} \times \vec{F}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 2 \\ 3 & 4 & -2 \end{vmatrix}$$

$$= \hat{i}(-2-8) - \hat{j}(-4-6) + \hat{k}(8-3)$$

$$= -10\hat{i} + 10\hat{j} + 5\hat{k}$$

Q4 (5)

$$I_1 = \frac{2}{5}M(2R)^2 = \frac{8}{5}MR^2$$

$$I_1 = \frac{1}{2}M(2R)^2 = 2MR^2$$

$$I_3 = \frac{M(2R)^2}{4} = MR^2$$

$$I_4 = \frac{M(2R)^2}{2} = 2MR^2$$

$$2(I_2 + I_3) + I_4 = x\ I_1$$

$$8MR^2 = x\frac{8}{5}MR^2$$

$$x = 5$$

Q5 (C)

Applying conservation of angular momentum

$MR^2\omega = (MR^2 + 2mR^2)\omega'$

$$\omega' = \frac{2M}{M+2m}$$

Q6 (B)

$K_{total} = K_{rotational} + K_{Translational}$

$K_{total} = \frac{1}{2} I_{cm}\omega^2 + \frac{1}{2} mV_{cm}^2$

$v_{cm} = R\omega$ for pure rolling

$I_{cm} = \frac{2}{5} mR^2$

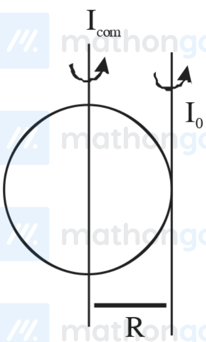
$K_{Rot} = \frac{1}{2} I_{cm}\omega^2 = \frac{1}{2} \times \frac{2}{5} mR^2 \times \frac{v_{cm}^2}{R^2} = \frac{1}{5} mv_{cm}^2$

$K_{Total} = \frac{1}{5} mv_{cm}^2 + \frac{1}{2} mv_{cm}^2 = \frac{7}{10} mv_{cm}^2$

$$\frac{K_{Rot}}{K_{Total}} = \frac{\frac{1}{5} mv_{cm}^2}{\frac{7}{10} mv_{cm}^2} = \frac{2}{7}$$

Q7 (A)

Solid sphere

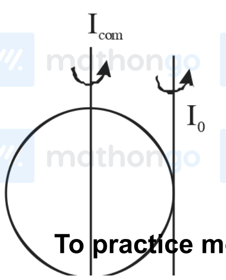


$I_0 = I_{com} + MR^2$ (Parallel Axis theorem)

$I_0 = \frac{2}{5} MR^2 + MR^2$

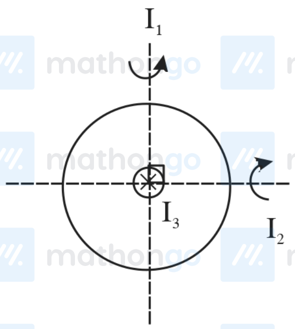
$I_0 = \frac{7}{5} MR^2$

Hollow sphere



$$I_0 = I_{com} + MR^2$$

$$= \frac{2}{3}MR^2 + MR^2 = \frac{5}{3}MR^2$$



$I_1 + I_2 + I_3$ (Perpendicular axis theorem)

By symmetry MOI

About 1" and 2" Axis are same i.e.

$$I_1 = I_2$$

$$\therefore\ 2I_1 = I_3 = MR^2\ \left(I_{com} = MR^2\right)$$

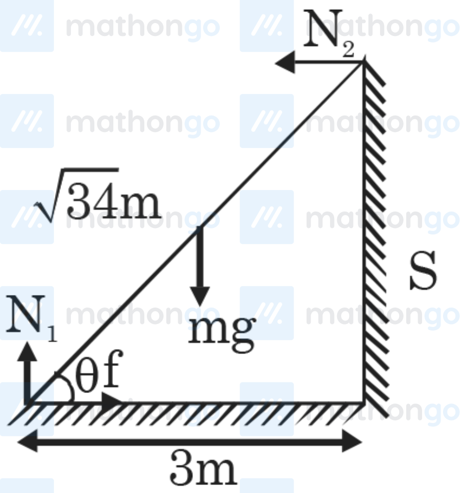
$$I_1 = \frac{MR^2}{2}$$

Similarly in disc

$$2I_1 = \frac{MR^2}{2}\left\{I_{com} = \frac{MR^2}{2}\right\}$$

$$I_1 = \frac{MR^2}{4}$$

Q8 (C)



$$f = N_2$$

$$N_1 = mg$$

$$N_2 \times \ell \sin\theta = mg\frac{\ell}{2}\cos\theta$$

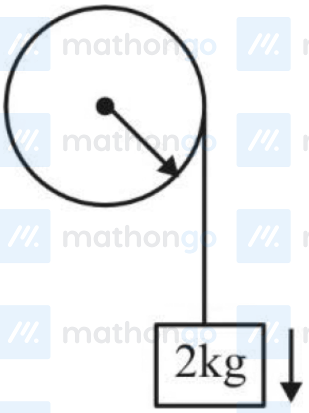
$$N_2 = \frac{mg}{2}\cot\theta$$

$$\frac{F_w}{F_f} = \frac{\frac{mg}{2}\cot\theta}{\sqrt{(mg)^2 + \left(\frac{mg}{2}\cot\theta\right)^2}}$$

$$= \frac{1}{\sqrt{1 + \frac{4}{\cot^2\theta}}}$$

$$= \frac{3}{\sqrt{109}}$$

Q9 (10)



$$2g - T = 2a \quad \ldots(1)$$

$$TR = \frac{MR^2}{2}\alpha \quad \ldots(2)$$

$$\alpha = \frac{a}{R} \quad \ldots(3)$$

$$T = 2a$$

$$2g - T = 2a$$

$$T = g = 10N$$

Q10 (C)

L_0 = angular momentum of shell about O.

As shell is rolling

so $V_{cm} = \omega R$

$L_0 = mV_{cm} R + I\omega$

$= 1 \times \omega R \times R + \frac{2}{3} R^2 \omega$

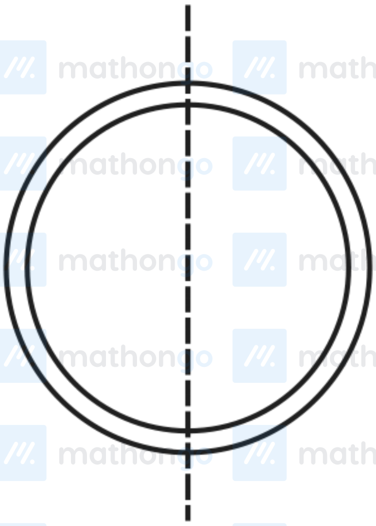
$= \frac{5}{3} R^2 \omega$

so a = 5

Q11 (8)

$$I_1 = \frac{ml^2}{3}$$

$$\ell = 2\pi r \Rightarrow \frac{\ell}{r} = 2\pi$$



$$I_1 = \frac{mr^2}{2}$$

$$\frac{I_1}{I_2} = \frac{2}{3}\left(\frac{\ell}{r}\right)^2$$

$$= \frac{2}{3} \times 4\pi^2 = \frac{8\pi^2}{3}$$

$$x = 8$$